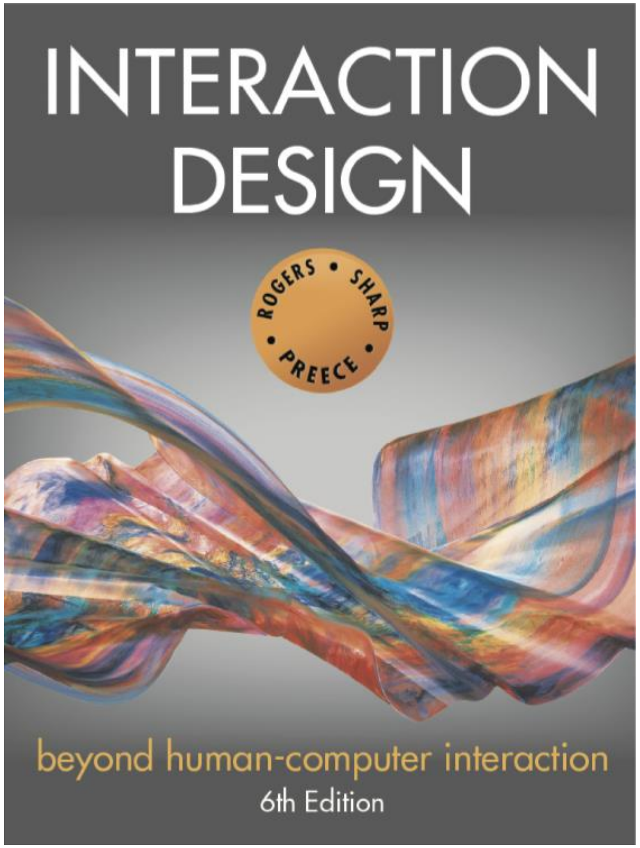


Chapter 6

EMOTIONAL INTERACTION

Overview

- Emotions and behavior
- Expressive and emotional design
  - How the ‘appearance’ of an interface can affect users
  - How technology can be designed for well being
- Affective computing and emotional AI
- Behavioral change
- Anthropomorphism

Emotions and behavior

- HCI has traditionally been about **designing efficient and effective systems**
- Now more about how to design interactive systems that make people respond in certain ways
  - For example, to be happy, to be trusting, to learn, or to be motivated
- Emotional interaction is concerned with how we feel and react when interacting with technologies
- Affective computing is improving with better recognition software and machine learning algorithms

Emotional Design

- The design of technology that can engender or cause desired emotional states or reactions in users.
- The focus is on how to design interactive products to **evoke certain kinds of emotional responses** in people
- https://www.interaction-design.org/literature/topics/emotional-design#:~:text=Emotional%20design%20is%20the%20concept,with%20products%2C%20brands%2C%20etc.

Emotional interaction

- Concerned with what makes us happy, sad, annoyed, anxious, frustrated, motivated, delirious, and so on
  - And translating this knowledge into design of different aspects of the user experience
- Why people become emotionally attached to certain products (for instance, virtual pets)
- Can social robots help reduce loneliness and improve well-being?
- How to change human behavior through the use of emotive feedback.
  - Emotive feedback aims to evoke emotional reactions in users.

Activity

- Try to remember the emotions you went through when buying a big ticket item online (for example, a refrigerator, a vacation, a computer)
- How many different emotions did you go through?

Activity (Cont.)

- Emotional Rollercoaster:
  - Thrill and anticipation of the product
  - Disappointment: It's too expensive
  - Annoyance and frustration: so many options to research and choose from
  - Relief: You choose the product
  - Tedious: payment process
  - Big sigh of relief: finally, the order is complete
  - Doubts: may the other product was better

Why has this simple way of obtaining visitor feedback been so effective?

Have you seen one of the feedback terminals shown in Figure at an airport after you have gone through security?

A Happyornot terminal located after security at Heathrow Airport

Happyornot Feedback Terminal

- Happyornot designed the feedback terminals that have been used in many airports throughout the world.
- The affordances of the large, colorful, buttons laid out in a semicircle, with distinct smileys, makes it easy to know what is being asked of the passerby, enabling them to select among feeling happy, angry, or something in between.
- More recent designs use a flat tablet as the interface.

Emotional Design in Advertisements

- Advertising agencies have developed a number of techniques to influence people's emotions.
- Examples include showing a picture of a cute animal or a child with hungry, big eyes on a website that "pulls at the heartstrings."
  – The goal is to make people feel sad or upset at what they observe and make them want to do something to help, such as making a donation.
- Following figure, for example, shows a web page that has been designed to trigger a strong emotional response in the viewer.

Pulling at the heart strings with an emotive message

A web page from Crisis (a UK homelessness charity)

Is it possible to design an interface to match or change how we are feeling?

- Should an interface be designed to improve how we feel?
  - If so, how?
- Our moods and feelings are continuously changing
  - How does the interface keep track and know when to do something?
- What moods match which kinds of interfaces?
- How would you design an interface for when someone is happy, angry, sad, bored, or focused?

How do emotions affect behavior and vice versa?

- Does being angry make you concentrate better or more distracted?
- When you are happy do you take more risks such as spend more money or buy more?
- Baumeister et al (2007) argue the relationship is more complex than a single cause-and-effect model

Automatic (affect) versus conscious emotions

- Emotions can be
  - **short-lived** (for instance, a fit of anger) or
  - **complex and long-lasting** (for example, jealousy)
- Emotions have been categorized as **automatic** or **conscious**:
- Automatic Emotions:
  - These happen rapidly, typically within a fraction of a second, and may dissipate just as quickly.
  - For example, getting startled by a sudden loud noise

Automatic (affect) versus conscious emotions (Reflect)

- Conscious Emotions:
  - These develop slowly and take a long time to go (for instance, reflection)
  - For example:
    - we can remain annoyed for hours when staying in a hotel room that has a noisy air conditioning unit.
    - An emotion like jealousy can keep simmering for a long period of time

Automatic (affect) versus conscious emotions (Reflect)

- Understanding how emotions work provides a way of considering how to design for user experiences that can trigger affect or reflect.

Designing User Experience

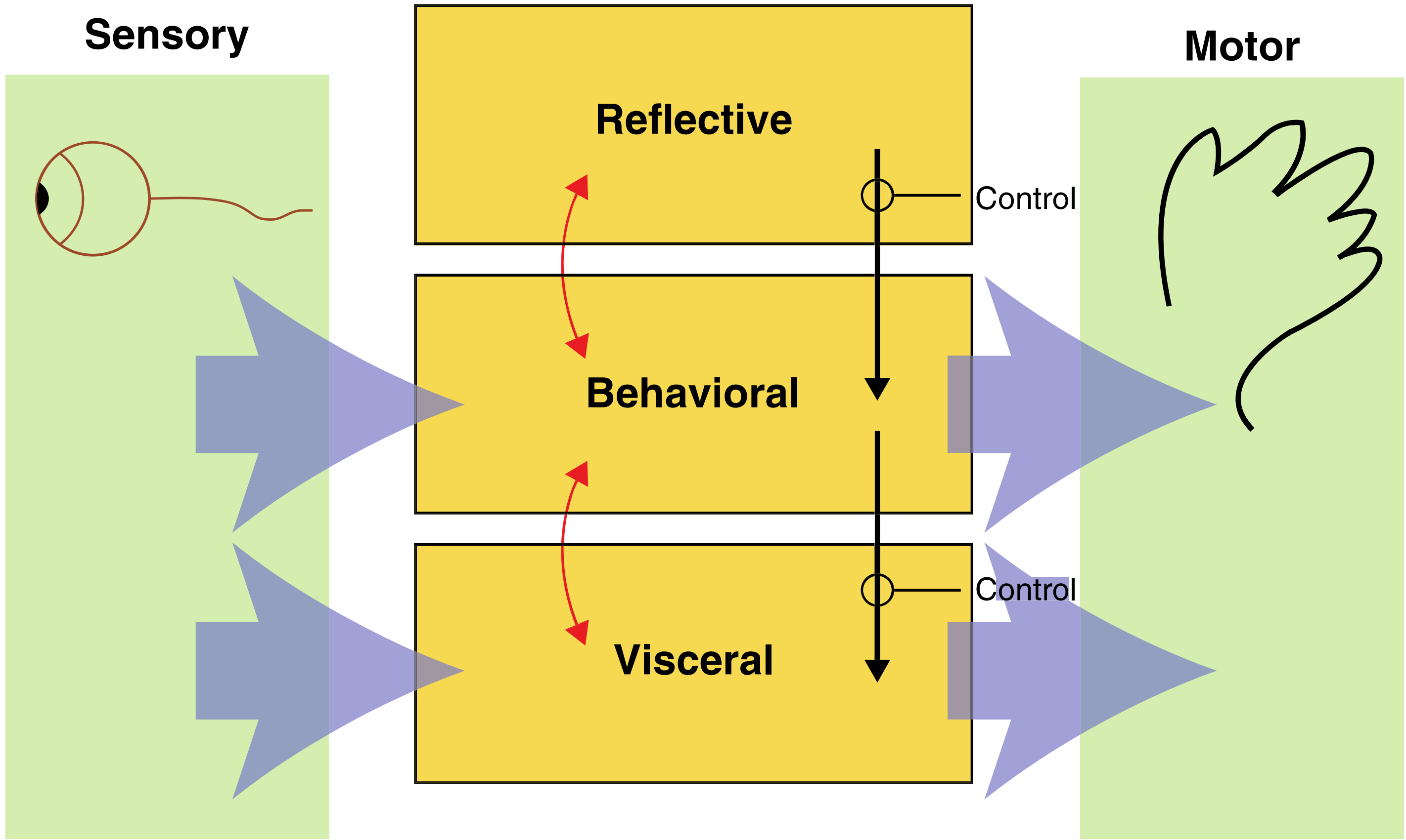
In a series of short videos, Kia Höök talks about affective computing, explaining how emotion is formed and why it is important to consider when designing user experiences with technology.

See www.interaction-design.org/encyclopedia/affective_computing.html

How emotions affect behavior and How behavior affects emotions

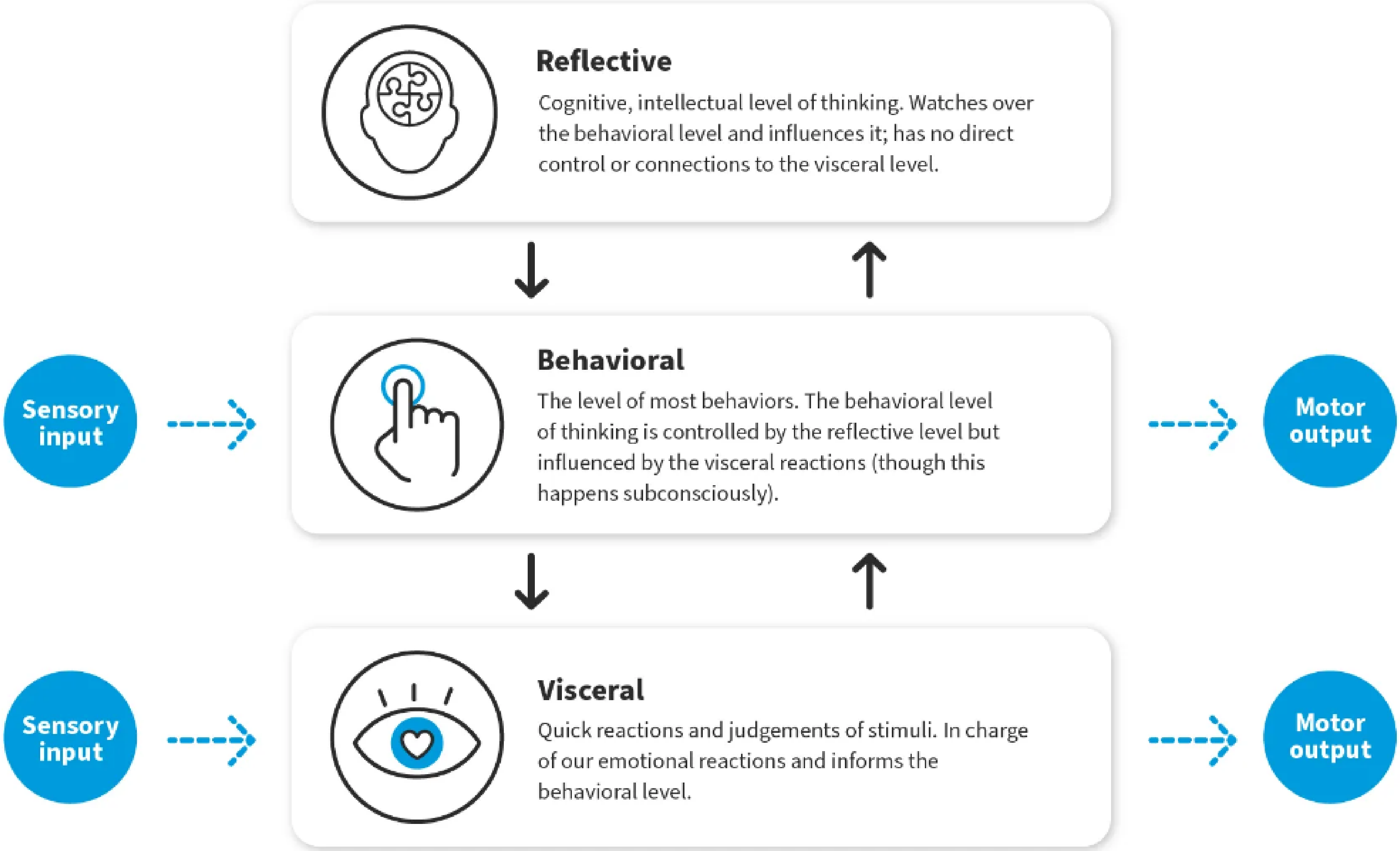
- Emotions affect behavior:
  - When people are happy, they typically smile, laugh, and relax their body posture.
  - When they are angry, they might shout, gesticulate, tense their hands, and screw up their face.
- Behavior affects Emotions:
  - A person's expressions can trigger emotional responses in others.
    - When someone smiles, it can cause others to feel good and smile back.

Ortony et al. (2005) model of emotional design



https://www.interaction-design.org/literature/article/norman-s-three-levels-of-design

Don Norman’s 3 Levels of Design



3 levels of emotional design

- **Visceral Level:**
  - At the lowest level are parts of the brain that are prewired to respond automatically to events happening in the physical world.
- **Behavioral Level:**
  - At the next level are the brain processes that control everyday behavior.
- **Reflective Level:**
  - At the highest level are brain processes involved in contemplating.

3 levels of emotional design

- The visceral level responds rapidly, making judgments about what is good or bad, safe or dangerous, pleasurable or abhorrent.
  - For example, many people will experience fear on seeing a very large hairy spider running across the floor of the bathroom, causing them to scream and run away.
- The behavioral level is where most human activities occur.
  - Examples include well-learned routine operations such as talking, typing, and swimming

3 levels of emotional design

- The reflective level entails conscious thought where people generalize across events or step back from their daily routines.
  - An example is switching between thinking about the story and special effects used in a horror movie and becoming scared at the visceral level when watching the movie.

Claims from model

- Our emotional state changes how we think
  - When frightened or angry, we focus narrowly and our bodies respond by tensing muscles and sweating
    - More likely to be less tolerant
  - When happy, we are less focussed and our bodies relax
  - We are more likely to overlook minor problems and be more creative

Designing with the three levels in mind

- Visceral design refers to making products look, feel, and sound good
- Behavioral design is about use, and it equates with traditional values of usability
- Reflective design is about considering the meaning and personal value of a product in a particular culture.

Application of Visceral Level Emotions in Design

- The initial impression a product makes based on its appearance, aesthetics, and sensory appeal.
- Designers can evoke the visceral level emotions by focusing on creating visually appealing and aesthetically pleasing interfaces.
- This involves attention to details such as color schemes, typography, graphics, and animations to evoke positive visceral reactions in users.

Application of Behavioral Level Emotions in Design

- The experience of using a product and the emotions that arise during interaction.
- It's about how well the product performs its intended function, how easy it is to use, and how satisfying the interaction is.
- At the behavioral level, emotions are influenced by factors such as usability, functionality, efficiency, and feedback.

Application of Behavioral Level Emotions in Design

- Designers can evoke the behavioral level of emotion by focusing on usability and user experience.
- This involves designing intuitive interfaces, minimizing cognitive load, providing clear feedback.
- By creating products that are easy to use and enjoyable to interact with, designers can foster positive emotional experiences

Application of Reflective Level Emotions in Design

- This level of emotion relates to the user's long-term relationship with the product and the meaning it holds for them.
- It's about the emotional connection users develop over time, as well as the personal and social significance of the product in their lives

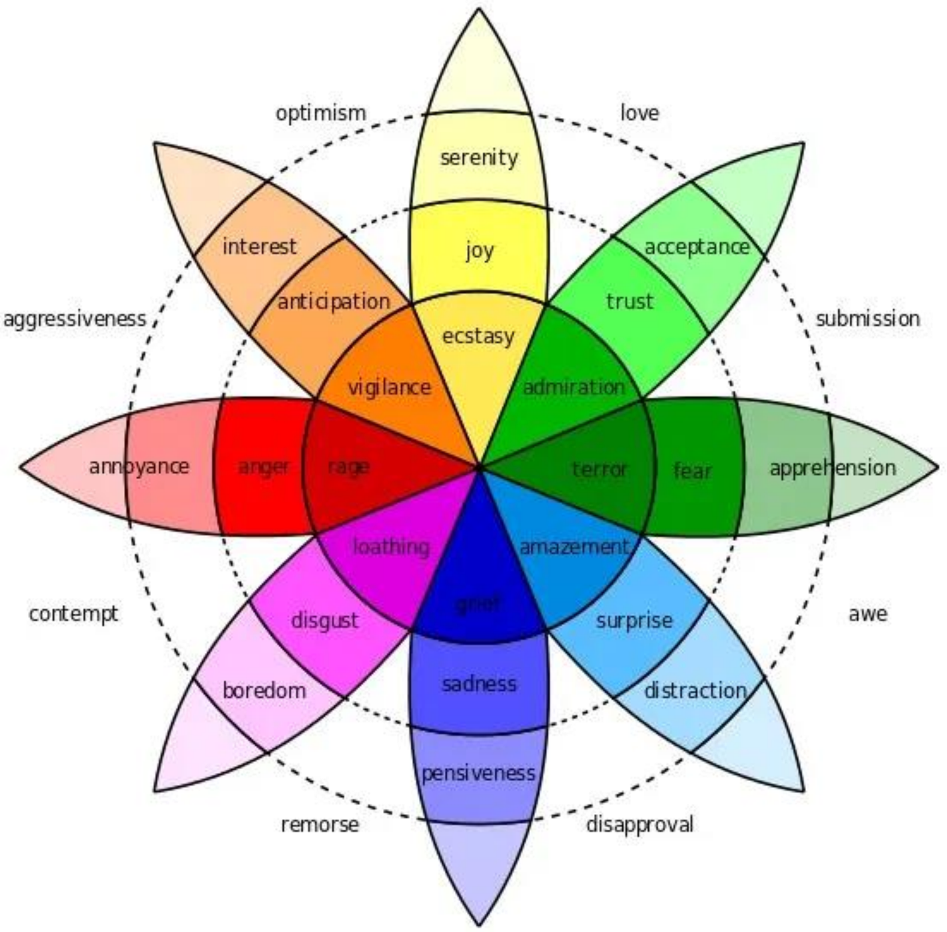
Application of Reflective Level Emotions in Design

- Designers can leverage the reflective level of emotion by creating products that resonate with users on a deeper emotional level.
- This involves designing for meaning and significance, aligning the product with users' values and aspirations
- By understanding the reflective level, designers can create products that evoke positive emotions and foster long-term relationships with users.

Analyzing a swatch watch design using the model

- Cultural images and graphical elements designed at the reflective level
- Affordances of use at the behavioral level
- Brilliant colors and wild design attract user's attention at the visceral level
- Emotional Design requires understanding who the target audience is and what the context of use will be

Plutchik's (1980) wheel of emotions



https://www.interaction-design.org/literature/article/putting-some-emotion-into-your-design-plutchik-s-wheel-of-emotions

Plutchik’s wheel

- Categorizes human emotions into 7 well known emotions
  - Anger, Disgust, Fear, Sadness, Anticipation, Joy and Surprise and Trust (not usually considered an emotion)
- Emotional labels are added to these
  - optimism, love, submission, awe, disproval, remorse, contempt, aggressiveness
- Colors used in the wheel reflect the intensity of an emotion
  - the darker the shade, the more intense the emotion is
- Can be used as a ‘colour palette’ akin to a UX mood board
  - elicit different kinds and levels of emotional response
- It does not provide prescriptive advice on how to instruct design for a selection of emotions

Plutchik's wheel of emotions

- Plutchik proposed that there are eight primary emotions, each with its own intensity levels, and that complex emotions arise from combinations of these primary emotions.
- The wheel is structured such that opposite emotions are placed directly across from each other, reflecting their contrasting nature.

Plutchik’s wheel of emotions

The primary emotions in Plutchik's Wheel of Emotions:

1. **Joy**: A feeling of happiness or delight.
2. **Trust**: A sense of confidence or belief in someone or something.
3. **Fear**: An emotional response to perceived danger or threat.
4. **Surprise**: A sudden or unexpected reaction to an event or stimulus.
5. **Sadness**: A feeling of sorrow, grief, or unhappiness.
6. **Disgust**: A strong aversion or revulsion towards something unpleasant or offensive.
7. **Anger**: A strong emotional response characterized by hostility, frustration, or irritation.
8. **Anticipation**: An expectation or eagerness about future events or outcomes.

Plutchik’s wheel of emotions

Each primary emotion on the wheel is positioned opposite its polar-opposite, representing the concept of bipolarity.

For example:

- Joy is opposite sadness.
- Trust is opposite disgust.
- Fear is opposite anger.
- Surprise is opposite anticipation

Plutchik’s wheel of emotions

Plutchik also proposed that there are secondary and tertiary emotions, which are blends or variations of the primary emotions.

These secondary and tertiary emotions arise from combinations of adjacent primary emotions on the wheel.

For instance, combining joy and trust may result in the emotion of love, while combining fear and surprise may result in the emotion of alarm.

Plutchik's Wheel

- Emotions on Plutchik's wheel may be combined as follows:
- Anticipation + Joy = Optimism (with its opposite being disapproval)
- Joy + Trust = Love (with its opposite being remorse)
- Trust + Fear = Submission (with its opposite being contempt)
- Fear + Surprise = Awe (with its opposite being aggression)
- Surprise + Sadness = Disapproval (with its opposite being optimism)
- Sadness + Disgust = Remorse (with its opposite being love)
- Disgust + Anger = Contempt (with its opposite being submission)
- Anger + Anticipation = Aggressiveness (with its opposite being awe)

Expressive interfaces

- A number of features have been developed to make an interface expressive,
  - **Visual techniques**: emojis, sounds, colors, shapes, icons, animations, videos, photos, and virtual agents.
  - **Other ways of conveying expressivity** include "sonifications" indicating actions and events (such as whoosh for a window closing, "schlook" for a file being dragged, or ding for a new email arriving) and vibrotactile feedback (such as distinct smartphone buzzes that represent specific messages from friends or family).

Expressive interfaces

- The motivation is often to

(1) create an emotional connection or feeling with people for instance, warmth, or sadness, and/or

(2) elicit certain kinds of emotional responses in people, such as feeling at ease, comfort, and happiness.

Expressive interfaces

- Provide reassuring feedback that can be both informative and fun
- Can also be intrusive, however, causing people to become annoyed and even angry
- Color, icons, sounds, graphical elements, and animations are used to make the 'look and feel' of an interface appealing
  - Conveys an emotional state
- In turn, this can affect the usability of an interface
  - People are prepared to put up with certain aspects of an interface (for instance, slow download rate) if the end result is appealing and aesthetic

Expressive interfaces

- Many websites, online shopping sites, and apps have been designed using a combination of these expressive features to good effect.
- Examples include retail sites like **Nike** and **Levis**, which have been creating aesthetic online shopping sites for many years using high-quality videos, emotive music, and striking images on their landing page

Design of website to be aesthetic

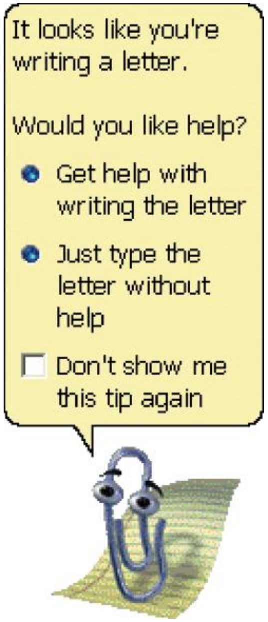
An image used on the landing page of Levis.com conveying coolness, sustainable materials, a grungy background, and aesthetic fonts

Aesthetic or Annoying?

- Sometimes expressive features, however, can turn out to be more **annoying** than **aesthetic**.
- Perhaps most well-known was Clippy, Microsoft's paperclip shaped Virtual Assistant that was included in Microsoft Office applications from Office 97 to Office 2003
- However, Clippy's intrusive and sometimes annoying presence, along with its tendency to appear at inconvenient times, led to widespread criticism and ridicule from users.

Microsoft’s Clippy and IKEA’s Anna

- Clippy was meant to be a helpful desktop assistant
- But was disliked by so many
  - annoying, distracting, patronizing
- Anna appeared as a virtual agent
  - Blinked, moved her lips and head to suggest facial expressions



(a)

(b)

(a) Microsoft’s Clippy and (b) IKEA’s Anna

The appearance of an interface

(a) Emotional icons were used in the 1980s to indicate rebooting or crashed computer

- Smiling apple face

(b) Nowadays, computers use more impersonal but aesthetically-pleasing icons to indicate that the user needs to wait

- Beachball

(a)

(b)

The design of thermostats

(a) The Nest thermostat has a minimalist and aesthetically-pleasing design

- Round face and simple dial
- Large font and numbers

(b) It is very different from earlier thermostat designs

- Utilitarian and dull

(a)

(b)

Affective Computing and Emotional AI

- *Affective computing is* concerned with how to use computers to recognize and express/convery emotions as humans do (Picard, 1998)
- It involves designing ways for people to communicate their emotional state
  - It uses sensing technologies to measure GSR(Galvanic Skin Response (GSR)), facial expressions, gestures, and body movement
- Explores how affect influences personal health
- *Emotional AI* aims to automate the measurement of feelings and behavior using AI to infer them from facial expressions and voice
- The goal is to predict user's emotions and aspects of their behavior
  - For example, what is someone most likely to buy online when feeling sad, bored, or happy

Techniques used

- Cameras for measuring facial expressions
- Biosensors placed on fingers or palms to measure GSR
- Affective expression in speech (for example, intonation, pitch, and loudness)
- Body movement and gestures using accelerometers and motion capture systems

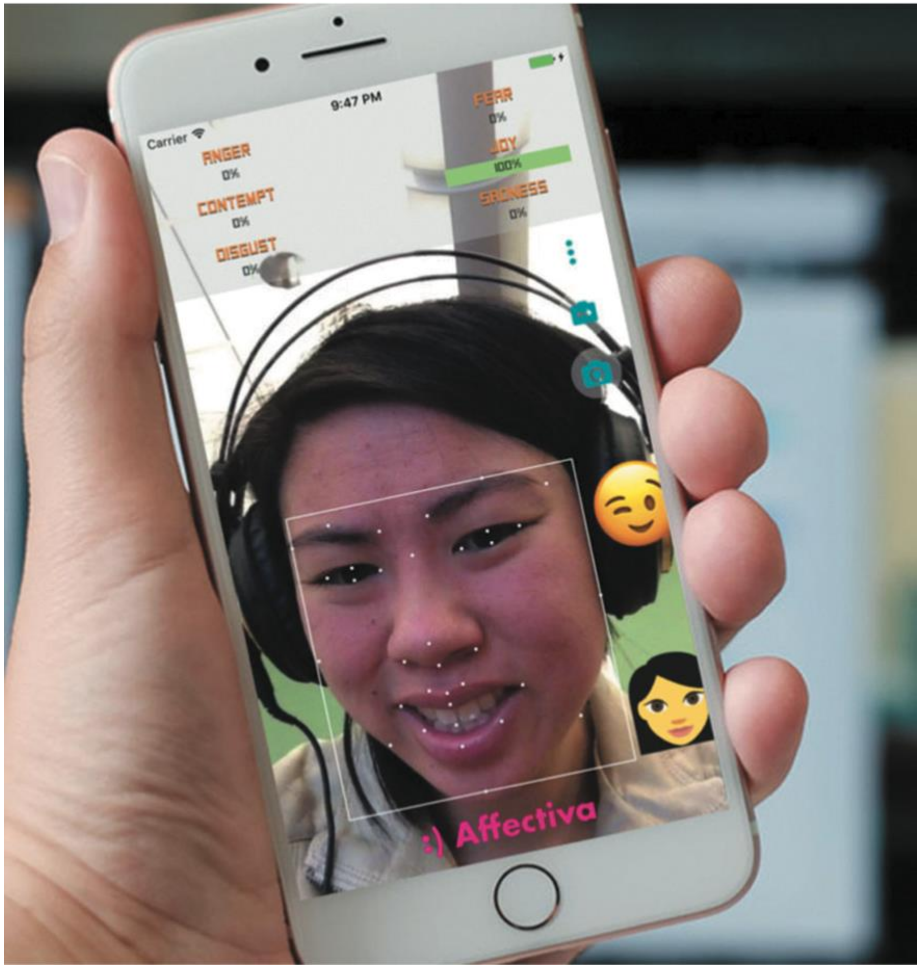
Classification of emotions

- Six core expressions typically measured:
  - Sadness, disgust, fear, anger, contempt, and joy
- Type of facial expression chosen by AI through detecting presence of absence of:
  - Smiling, eye widening, brow raising, brow furrowing, raising a cheek, mouth opening, upper-lip raising, and wrinkling of the nose

Classification of emotions

- If a person furrows their brow and wrinkles their nose (i.e., screws their face up) when an ad pops up, this suggests that they feel disgust,
- whereas if they start smiling, it suggests that they are feeling happy.
- The website can then adapt its ad, movie storyline, or content to what it perceives the person needs at that point in their emotional state.

Facial coding using Affdex software



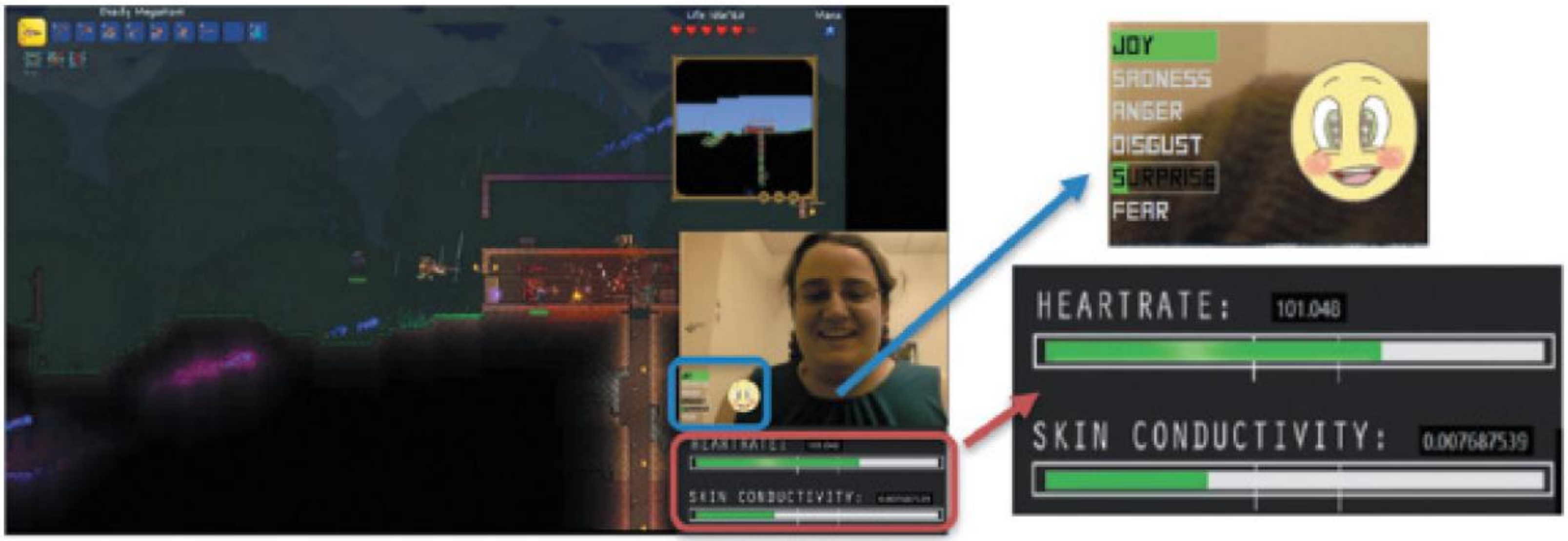
How is this emotional data used?

- If user screws up their face when an ad pops up
  ➡ Feel disgust
- If user starts smiling
  ➡ They are feeling happy
- **Website** can adapt its ad, movie storyline, or content to match user’s emotional state
- If system used in a **car**, it might detect an angry driver and suggest they take a deep breath
- Eye-tracking, finger pulse, speech, and words/phrases also analyzed when tweeting or posting to Facebook

Indirect emotion detection

- Also used to infer or predict someone's behavior
  - e.g, determining a person's suitability for a job or how they will vote in an election
- Do you think it is ethical that technology can read your emotions from your facial expressions or from your tweets?

All the Feels app: Player's emotions



All the Feels app showing the biometric data of a streamer playing a video game

Detecting and Reflecting on Moods

- How can technology be designed to help people understand more about their moods and mood swings?
- We can be in a good mood one day and then a bad mood the next day. How does this happen and why?
- Not possible to detect different moods in the way tech is used for emotion detection
  - often not expressed through obvious physiological responses
- Mobile mood tracker apps (e.g., Moodnotes, Daylio)
  - intended to help people keep track of their moods
  - to reflect more on why they might be feeling gloomy or cheerful
  - understanding their moods is assumed to help improve well-being
- Virtual reality is also used to enable people to explore their moods (e.g. Mood Worlds)

Mood Worlds: A VR exploratory tool

Nadine Wagener et al (2022)

A participant using the VR app Mood Worlds to visualize and explore their emotions

Behavioral change

- Interactive computing systems designed to change people's attitudes and behaviors (Fogg, 2003)
- A diversity of techniques now used to change what they do or think
  - Pop-up ads, warning messages, reminders, prompts, personalized messages, recommendations, or Amazon 1-click
  - Commonly referred to as *nudging*

Nintendo's Pocket Pikachu

Developed to change bad habits and improve well being

- Designed to motivate children to be more physically active on a regular basis
- Owner of the digital pet that 'lives' in the device is required to walk, run, or jump
- If owner does not exercise, the virtual pet becomes angry and refuses to play anymore

How effective?

- Can interactive technologies that monitor, nag, or behave like a human keep them interested in looking after it and in doing so make themselves more fit?
- How does looking after a virtual pet change a child's behavior?
  - Emotional attachment
  - Happy Pokemon makes them feel good
  - Sulking Pokemon makes them feel bad

Dilemma: Should voice assistants teach kids good manners?

- Many children talk to Alexa as if she was their friend
- They also learn that it is not necessary to say please and thank you to her when asking questions
- Is this lack of using etiquette a problem?
- Would it transfer over to real life situations?
  - For example, demanding "Auntie, get me my drink."
- Should parents or voice assistants teach their kids good manners
- How much parental control should voice assistants be given?
  - Would children find it weird or creepy that their Alexa (who is their friend) nags them to clean their teeth?
- Recent research has shown that children know to treat humans differently compared with the way they talk to a voice assistant (Alexis Hiniker et al., 2021)

Sustainable HCI

- Focus on designing tech interventions to help people reduce their energy consumption
- An effective technique is to provide homeowners with feedback on their consumption
- Simple infographics and emoticons are often most powerful:
  - Encourage people to reflect and see what they can change to reduce their energy consumption
- Peer pressure and social norms are also powerful methods

Stevie the robot entertaining residents while at a retirement home

Commercial Social Robots

- A number of commercial physical robots have been developed specifically to support care giving for older adults.
- Early ones were designed to be about 2 feet tall and were made from white plastic with colored parts that represented clothing or hair, often having big eyes and holding a tablet to display messages.

Summary

- Emotional aspects of interaction design are concerned with how to facilitate certain states (for example, pleasure) or avoid reactions (for instance, frustration)
- Well-designed interfaces can elicit good feelings in people
- Aesthetically-pleasing interfaces can be a pleasure to use
- Expressive interfaces can provide reassuring feedback to users
- Badly designed interfaces make people frustrated, annoyed, or angry
- Emotional AI and affective computing use AI and sensor technology for detecting people’s emotions by analyzing their facial expressions and conversations
- Emotional technologies can be designed to persuade people to change their behaviors
- Anthropomorphism is the attribution of human qualities to objects
- Increasingly, robots are being used as companions in the home